**HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY**
**School of Information and Communication Technology**







# PROJECT REPORT

## Customer Churn Prediction and Segmentation for a Digital Bank

**Course:** Intro To Business Analyst
**Class ID:** 175884
**Supervisor:** Ph.D Bui Quoc Trung



| Member | Student ID |
|---|---|
| **Hoang Trong Tan** | 202416751 |
| **Vu Thuong Tin** | 20230091 |
| **Tran Quang Huy** | 202416706 |
| **Nguyen Le Duc Anh** | 20235472 |
| **Nguyen Duc Dung** | 202416681 |



**Ha Noi, 2026**

# Contents

# List of Figures

# List of Tables

# List of abbreviations

| | |
|---|---|
| A/B | Randomized comparison of a treatment arm against a control or alternative arm |
| API | Application Programming Interface |
| ARI | Adjusted Rand Index |
| CI | Confidence interval |
| CRM | Customer Relationship Management |
| K-Means | K-means clustering algorithm |
| KPI | Key Performance Indicator |
| MDE | Minimum Detectable Effect |
| MLOps | Model deployment, monitoring and governance practices |
| OOD | Out-of-distribution |
| PCA | Principal Component Analysis |
| PR-AUC | Area under the precision–recall curve |
| ROC-AUC | Area under the receiver operating characteristic curve |
| SLA | Service Level Agreement |

# 1 Introduction

**Churn is the loss of a relationship, not merely a closed account.** Retail banking is built on repeated financial decisions: where customers receive income, make payments, hold savings, seek credit, and obtain help when something goes wrong. A customer may retain a current account while moving these routines elsewhere. Churn therefore represents a deeper commercial loss than a single product closure. It removes future opportunities to serve the customer, weakens the deposit and payment relationship, and can expose an unresolved weakness in the customer journey. The operational question is consequently broader than "who left?" A bank needs to know which relationship is weakening, which customer need is visible, which team owns the response, and which action can be delivered without creating further friction.

**Digital banking has made retention an urgent operating decision.** The World Bank reports that 79% of adults worldwide had a financial account in 2024, up from 74% in 2021. Its *Global Findex 2025* evidence base covers about 148,000 adults across 141 economies and tracks the growing connection among financial access, payments, saving, borrowing, mobile connectivity, and digital safety (**??**). As everyday banking moves onto mobile and online channels, customers can compare providers, open additional relationships, and shift routine activity with far less effort than in a branch-centred model. The commercial pressure is visible in Baringa's 2025 survey of 4,000 retail-banking customers in the United Kingdom and the United States: more than one third had changed provider during the preceding five years, while 62% said that a better digital experience would motivate them to switch (**?**). These figures place customer experience alongside price and product as a central retention concern.

**The project data show a material churn burden.** The verified project cohort contains 10,000 customers, of whom 2,037 carry the historical `Exited` label: a prevalence of 20.37% `OBSERVED_STATIC_DATA` (**?**). Put plainly, roughly one in every five customer relationships in the available cohort ended. That scale makes indiscriminate outreach a poor response. A bank that treats all 2,037 customers as identical risks spending scarce CRM capacity on the wrong journey, offering a promotion when service recovery is needed, or repeatedly contacting customers who have already disengaged. The useful unit of work is therefore an eligible, prioritised customer group with a clear operational owner, a channel, a service or product action, a cost ceiling, and an outcome to measure.

**Customer exit follows the quality of the relationship customers actually experience.** The relevant signals extend beyond balances and demographics. Payment failure, difficult onboarding, unclear fees, delayed complaint resolution, inaccessible support, weak security confidence, and a frustrating digital journey can each alter how customers use a bank. Longitudinal evidence from a European financial-services company illustrates the analytical depth required: Van den Poel and Larivière studied 47,157 customers using relationship history, behavioural variables, demographics, and macro-environmental conditions, and identified critical churn periods early in the customer relationship (**?**). For a digital bank, the implication is practical. Retention work must connect product telemetry, service cases, transaction behaviour, contact permission, and customer outcomes, then distinguish a service-recovery case from a routine engagement opportunity. A complaint should trigger resolution; it should

not be interpreted as a cue for a generic sales offer.

**Prediction becomes valuable only when it improves the next decision.** Customer-churn analytics should help a bank allocate finite attention. A score can prioritise review; behavioural segmentation can make the proposed action more relevant; value rules can set a permissible cost; and eligibility rules can suppress customers with no consent, excessive contact frequency, or an unresolved case. None of these components substitutes for the others. A high-risk score without an action creates a queue, not a retention strategy. A segment without a distinct treatment becomes a descriptive label. A campaign without a control group cannot establish whether it changed retention. The strongest operating design links each proposed contact to a reason code, a responsible owner, a delivery channel, a budget, and a measurable customer and business outcome.

**This report develops that decision system for a digital-bank setting.** It begins with the verified static Kaggle customer extract and uses `Exited` as the historical outcome available to the project. A Python-generated six-month behavioural layer, labelled `SYNTHETIC_BEHAVIOR`, provides the event structure needed to design customer features, behavioural segments, staff workflows, and monitoring controls. The report then brings classification, segmentation, eligibility, economics, and experimentation into one decision framework. Its central question is deliberately operational: *which customers merit attention, what relationship pattern is visible, which intervention fits that pattern, through which channel, at what cost, and what result would justify scale?* The resulting product is a staff-facing retention decision tool that combines customer intelligence with consent, service, capacity, and measurement discipline.

# 2 Banking background and decision context

## 2.1 Customer relationship, product relationship, and churn are different objects

In a digital retail bank, the customer lifecycle runs from acquisition and identity verification through funded account, first valuable transaction, routine use, product deepening, declining engagement, dormancy, exit, and sometimes reactivation. The same person can own several products. Closing a card is not necessarily leaving the bank; a funded but unused account is not necessarily an active relationship; and a reduced balance is not automatically balance flight. These distinctions are not semantic niceties: they define the unit of analysis and prevent the dashboard from promising more than the data can support.

Table 2 states the project taxonomy. Only the first row is represented by the source label. The other concepts require event history, product records, or intervention data that the current base does not contain.

Table 2: Churn and relationship taxonomy used in this report [scope control].

| Concept | Operational meaning | Evidence available here |
|---|---|---|
| Customer attrition / churn | Relationship exit represented by the supplied label | Historical static proxy only (`OBSERVED_STATIC_DATA`) |
| Account closure | Closure of a particular account | Not observed |
| Product attrition | Loss of a card, deposit, or loan product | Not observed |
| Dormancy | No qualifying activity in a declared window | Demonstration only in simulated events (`SYNTHETIC_BEHAVIOR`) |
| Digital disengagement | Decline in app or online use relative to a personal baseline | Demonstration only in simulated events (`SYNTHETIC_BEHAVIOR`) |
| Balance flight | Sustained material decrease in deposits | Not inferable from one balance snapshot |
| Preventable churn | Churn avoided because of an intervention | Unknown until a controlled experiment (`EXPERIMENTALLY_MEASURED`) |

## 2.2 The operational decision, not just the model

The intended decision cycle is: (1) check contact permission, fatigue cap, and policy exclusions; (2) route an unresolved service case to service recovery rather than promotion; (3) assign a behavioural segment using the persisted segment rule; (4) overlay a validated propensity tier and an explicitly governed value tier when those components exist; (5) allocate eligible customers to a treatment or control condition; and (6) record exposure, outcome, cost, complaint, opt-out, and the reason for any human override. The no-contact decision is a valid outcome, not a missing action.

This design separates four constructs that are often conflated. Behavioural segmentation groups comparable observed or simulated usage patterns. A churn-propensity tier is a supervised statistical estimate. A value tier requires contribution data or transparent scenarios; balance and estimated salary are not profit. Treatment eligibility reflects consent, open cases, suitability, and capacity. A customer should never receive an offer merely because they fall into a cluster or have a high score.

## 2.3 Economics and causal discipline

Retention economics should be framed in incremental terms. Where real contribution data are unavailable, the analysis may use a labelled scenario but cannot call it realised revenue. For a planned campaign, the relevant relationship is

$$\text{Incremental net value} = N_{\text{eligible}} \times \Delta r \times V_{\text{saved}} - N_{\text{contacted}} \times C_{\text{contact}} - C_{\text{incentive}} - C_{\text{execution}} - C_{\text{experiment}}, \tag{1}$$

where $\Delta r$ is the incremental retention effect and $V_{\text{saved}}$ is incremental contribution per saved customer. In the current project, both are unknown. Equation 1 is therefore a `ASSUMPTION_SCENARIO` template, not a profit estimate. If $V_{\text{saved}} > 0$, the break-even effect is total incremental campaign cost divided by

$N_{\text{eligible}} \times V_{\text{saved}}$. This makes the decision testable: Finance can set the approved assumptions before launch; Analytics can later replace them with an experiment estimate and its uncertainty interval.
The causal boundary is central. A high historical churn rate in a cluster may prioritise further investigation, but it does not prove that the cluster's proposed treatment will retain customers. Randomization within a frozen, eligible segment is needed to separate the treatment outcome from background change, self-selection, seasonality, and regression to the mean. Promotional outreach is suppressed when service recovery is more appropriate.

### 2.4 Stakeholder decisions and safeguards

The same analytical output serves different decisions. The executive owner needs a portfolio priority and a review date. CRM needs a contactable population, channel, suppression rules, and capacity. Product and Operations need a falsifiable friction hypothesis and an instrumentation backlog. Finance needs a budget cap and break-even sensitivity. Risk and Compliance need aggregate fairness checks, consent controls, an audit trail, and a stop authority. Customers need transparent service choices and an opt-out pathway, not disclosure of an internal score. This report keeps gender and recorded country out of segment fitting and treatment selection; they can only support aggregate fairness review once a legitimate, governed use has been defined.

## 3 Data overview, provenance, and analytical design

### 3.1 Observed static base and source audit

The observed base is the course-specified Kaggle extract, *Bank Customer Churn Dataset*, uploaded by Gaurav Topre (**?**). Preprocessing version `kaggle-static-preprocess-v1` read 10,000 rows and retained all 10,000: no row was deleted, no value was imputed, and the audit found zero duplicate customer identifiers and zero missing values in the retained fields. The raw-file SHA-256 recorded in the quality manifest is `90f6625c31bdfcb0cde3b94c4f3004e43795e382921d0a5be67c4bd490b38818`. This checksum demonstrates which local file was used; it does not validate the upstream collection process or representativeness.
The dataset contains one row per customer and a static historical label. It includes customer ID, credit score, recorded country category, gender category, age, tenure, balance, number of products, card flag, active-member flag, estimated salary, and `Exited`/`churn`. The preprocessing pipeline separates the target into a label table, so clustering and feature-generation code cannot read it by accident. Its overall historical label prevalence is 2,037/10,000 = 20.37% (95% Wilson CI: 19.59%–21.17%). This is a descriptive sample statistic, not an expected future bank churn rate.
The exact Kaggle URL is now recorded in the bibliography; however, the repository does not yet contain a documented licence review or source-version metadata from Kaggle. That provenance item remains **BLOCKED**. Moreover, the source currency for balance and estimated salary is unspecified. The report never treats either field as salary income paid into the bank, deposit value, revenue, customer lifetime value, or a targeting rule.

Table 3: Static source contract and handling [OBSERVED_STATIC_DATA].

| Element | Contract and limitation |
|---|---|
| Grain | One row per pseudonymous source customer; unique, nonblank identifier required. |
| Fields | Demographic/profile fields, product count, balance, estimated salary, active-member flag, and static historical churn label. |
| Quality outcome | 10,000 read; 10,000 retained; 0 duplicates; 0 retained nulls; 0 rows dropped. |
| Target boundary | churn is stored separately from model-ready customer attributes. |
| Time semantics | No observation date, event history, prediction point, or label window. A next-quarter claim is unsupported. |
| Sensitive/proxy fields | Gender and country are excluded from behavioural clustering and proposed treatment selection. |
| Financial semantics | Balance and estimated salary use unspecified source currency; neither is contribution or CLV. |

## 3.2 Why a synthetic behavioural extension is used

The static base has no transaction, app, channel, or service-case history. It cannot answer whether a customer was recently active, whether app use is declining, whether a payment failed, or whether an unresolved case should suppress an offer. To rehearse those analytical and engineering requirements, the project generates six monthly periods of labelled SYNTHETIC_BEHAVIOR data using a fixed seed. The generator receives only label-free static attributes and seeded random variation. It explicitly rejects an input table that contains churn, so the primary behavioural simulation is not label-conditioned. The resulting tables model a minimal customer-event architecture: customer-month balance-flow scenarios, transaction-month channel and type facts, digital-month app activity and failed-payment scenarios, simulated service cases, a cutoff-safe feature snapshot, and a segment assignment. The generator verifies grains and basic plausibility: exactly six customer-month rows per customer, non-negative transaction counts and amounts, app-active days in 0–31, sessions no fewer than active days, valid case timelines, and a unique feature row per customer. A synthetic incoming-flow event is specifically not interpreted as an observed salary payment; a synthetic loan repayment is not evidence of actual loan ownership.

Table 4: Canonical analytical tables and provenance [OBSERVED_STATIC_DATA + SYNTHETIC_BEHAVIOR].

| Table | Grain | Purpose and source status |
|---|---|---|
| customer_base | one customer | Static profile; OBSERVED_STATIC_DATA. |
| customer_snapshot | customer × simulated month | Simulated inbound/outbound-flow scenario; SYNTHETIC_BEHAVIOR. |
| transaction_month | customer × simulated month × channel/type | Simulated count and amount; SYNTHETIC_BEHAVIOR. |
| digital_month | customer × simulated month | Simulated active days, sessions, feature, failed payment; SYNTHETIC_BEHAVIOR. |
| service_case | simulated case | Simulated open/resolved lifecycle and category; SYNTHETIC_BEHAVIOR. |
| feature_snapshot | customer × cutoff | Features computed using months at or before cutoff; SYNTHETIC_BEHAVIOR. |
| segment_assignment | customer × segment version | Label-free behavioural grouping; SYNTHETIC_BEHAVIOR. |
| risk_prediction | customer × model version | Planned only; no trained artifact exists yet. |

The extension supports method development and interface testing, not discovery of actual customer motives. Its correlations are construction assumptions. Changing the seed changes simulated customers; it does not constitute external validation. Every display below marks whether its numbers originate in the static base, the simulation, a future model estimate, or a scenario.

## 3.3 Analytical workflow and controls

The workflow deliberately separates descriptive analysis, segmentation, prediction, and treatment measurement. First, static fields support a source audit and historical cohort descriptions. Second, simulation produces event facts and cutoff-safe behavioural features. Third, clustering is fit without the churn label, IDs, gender, or country. Only after assignment is frozen is the historical static label joined for a descriptive cross-tab. Fourth, a later classification pipeline must use train-only transformations, untouched test data, calibration, subgroup diagnostics, and capacity-based threshold selection. Finally, treatment performance is measured by a future randomized pilot, not imputed from a churn score.

This sequence controls several common errors: target leakage into clustering, learning transformations from the holdout, confusing synthetic behaviour with observation, calling balance a value metric, and converting association into causal explanation. It also creates a practical audit trail: every table declares grain; each derived file has a version and source tag; and user-facing recommendations can cite the artifact that produced them.

# 4 Customer analysis and behavioural segmentation

## 4.1 Analytical question and evidence status

The completed customer-analysis module asks a narrow, decision-relevant question: *within a six-month synthetic behavioural rehearsal, are there stable, operationally distinct usage patterns that justify different service and experiment hypotheses?* It does not ask which customers will certainly churn, which customer has a particular need, or which treatment will create retention. The answer is based on a V3 K-Means baseline fitted to all 10,000 simulated feature snapshots and then overlaid with the static historical label for description only.

Nine behaviour-only features enter the baseline: six-month transaction count and amount, app-active days, failed payments, simulated inbound flow and number of inflow months, cash-withdrawal share, an inflow trend ratio, and unresolved-case share. Positive skewed counts and amounts are transformed with `log1p`; the feature matrix is standardized before K-Means. The churn label is absent from the generator, feature construction, and cluster fitting. Segment IDs are then made interpretable by labelling the group with higher mean app-active days as `segment_high_digital`; this is a neutral behavioural description, not a value judgement.

The V3 baseline produces a silhouette value of 0.373. Across five full generator-plus-clustering runs with seeds 11, 23, 37, 53, and 71, the high-digital segment share lies between 52.46% and 52.70%; adjusted Rand index relative to seed 11 ranges from 0.875 to 0.885 for the other four runs. These diagnostics show that this particular simulated construction gives a repeatable two-group split. They do not prove that two true customer personas exist at a real bank. The current V3 artifact also lacks a documented comparison across adjacent $k$ values and alternative clustering algorithms; therefore $k = 2$ is a transparent operational baseline, not a final scientific optimum. That gap must be closed before a production-like segmentation gate can pass.

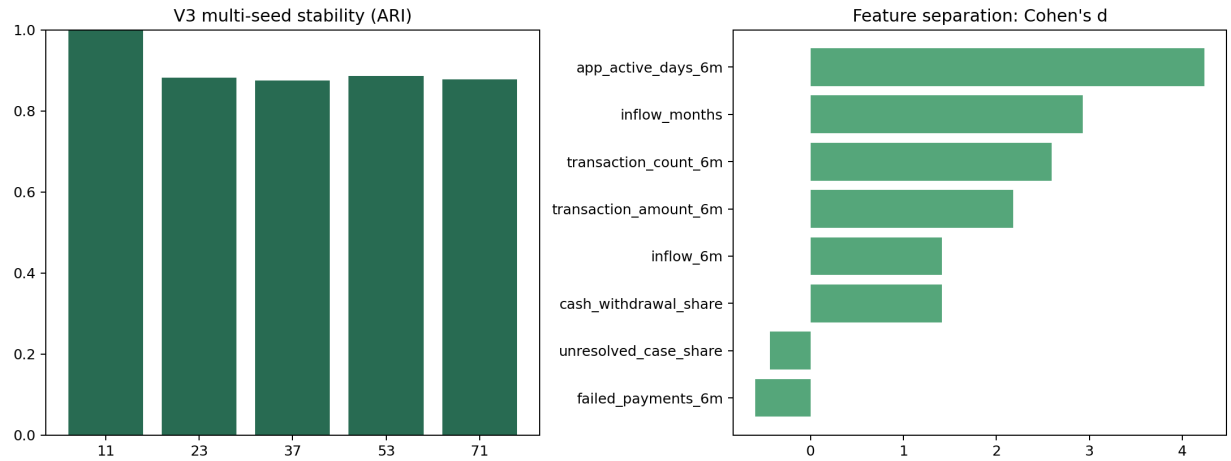



Figure 1: [SYNTHETIC_BEHAVIOR] V3 five-seed stability diagnostic and behaviour-feature separation. The chart assesses repeatability of the simulated pipeline, not validation of real bank personas.

## 4.2 Portfolio-level result: two different behavioural patterns

The two groups are operationally large and almost balanced. The high-digital-routine segment contains 5,285 customers (52.85%); the lower-digital-relationship segment contains 4,715 (47.15%). Ta-

ble 5 reports the cluster profiles. All rows except historical-label columns are derived from the synthetic extension. The strongest separation is app-active days: 67.17 versus 24.42 days over six simulated months (Cohen's $d = 4.24$). The high-digital group also has more inflow months and more simulated transactions. The lower-digital group has more failed payments and a larger unresolved-case share.

Those differences support differentiated operational hypotheses. They do not establish customer satisfaction, financial distress, the reason for a failed payment, or the cause of churn. In particular, the cash-withdrawal-share result is an artefact of the present generator: it is identical at one-third in both groups while the reported standardized difference is distorted by negligible within-group variance. It should not be used in a segment narrative or treatment rule until the feature definition and diagnostic are corrected. Surfacing this anomaly is preferable to silently using an attractive but unstable feature.

Table 5: V3 behavioural segment profile, six simulated months; historical label shown only as a post-fit overlay [SYNTHETIC_BEHAVIOR + OBSERVED_STATIC_DATA].

| Measure | High digital routine | Lower digital relationship | Evidence class |
|---|---|---|---|
| Customers, share | 5,285; 52.85% | 4,715; 47.15% | SYNTHETIC_BEHAVIOR assignment |
| App-active days, 6m (mean) | 67.17 | 24.42 | SYNTHETIC_BEHAVIOR; $d = 4.24$ |
| Inflow months, 6m (mean) | 4.83 | 1.35 | SYNTHETIC_BEHAVIOR; $d = 2.93$ |
| Transaction count, 6m (mean) | 41.71 | 26.35 | SYNTHETIC_BEHAVIOR; $d = 2.60$ |
| Transaction amount, 6m (mean) | 1,415.51 | 824.41 | SYNTHETIC_BEHAVIOR; source currency unspecified; $d = 2.18$ |
| Failed payments, 6m (mean) | 0.25 | 0.64 | SYNTHETIC_BEHAVIOR; $d = -0.59$ |
| Unresolved-case share (mean) | 7.28% | 20.91% | SYNTHETIC_BEHAVIOR; $d = -0.44$ |
| Static historical churn label | 751/5,285 = 14.21% | 1,286/4,715 = 27.27% | OBSERVED_STATIC_DATA post-fit cross-tab |
| 95% Wilson CI for label rate | 13.29%–15.18% | 26.02%–28.56% | OBSERVED_STATIC_DATA post-fit cross-tab |

The observed-static cross-tab differs by 13.06 percentage points: the lower-digital group has a historical label rate about 1.92 times that of the high-digital group. This is a useful prioritisation signal for an exploratory pilot design, subject to the source limitations. It is not a forecast, an individual probability, or evidence that low app activity caused exit. The simulation itself is conditioned on static fields such as the recorded active-member flag, age, product count, and estimated salary; it therefore

cannot be used to discover a causal engagement mechanism that has already been embedded in the generator.

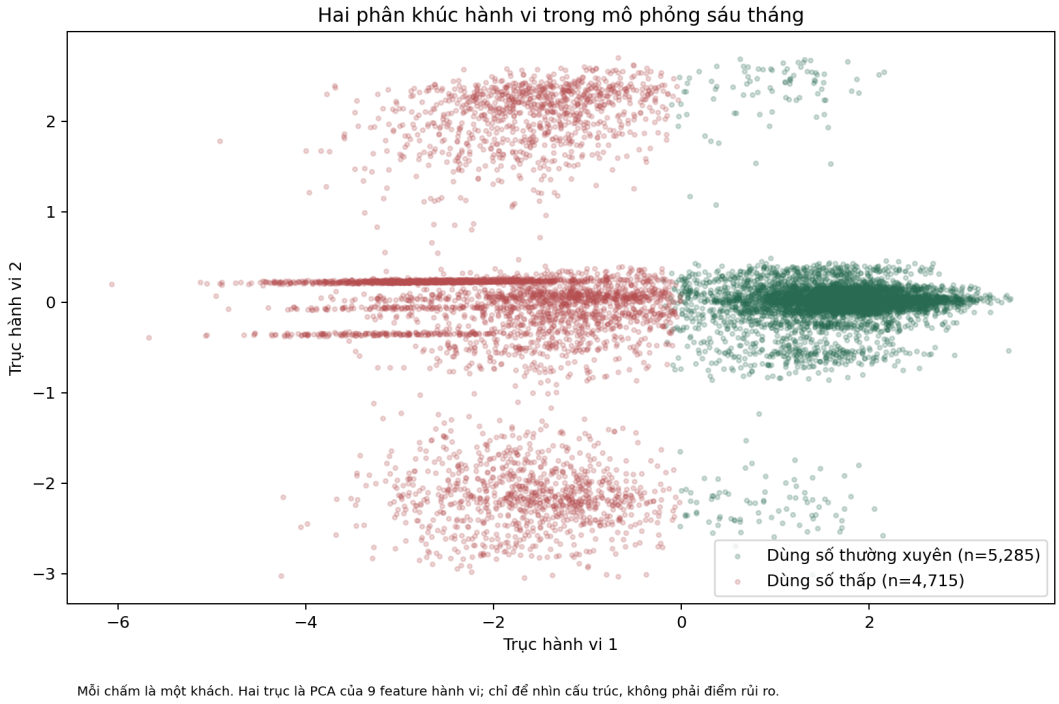



Figure 2: [SYNTHETIC_BEHAVIOR] Two V3 behavioural groups in a PCA projection of the simulated feature space. The projection is a visual summary only; it is neither a risk score nor a treatment recommendation.

## 4.3 Concrete persona dossiers: Linh and Minh

To make the segment cards usable by CRM, Product, and Care, the project gives each behavioural pattern a synthetic persona handle. **Linh** and **Minh** are not customers, not representative individuals, and not customer records. Their names make the operating distinction memorable; every behavioural detail below remains a group-level SYNTHETIC_BEHAVIOR result, while the profile fields and historical label cross-tabs are OBSERVED_STATIC_DATA. Gender, recorded country, estimated salary, and balance are never used to choose a channel, offer, price, or eligibility.

### 4.3.1 Persona A — Linh: regular digital self-service relationship

**Who Linh represents.** Linh represents 5,285 customers (52.85% of the portfolio). In the static base, this group has mean age 39.6, mean credit score 653.2, 44.4% recorded female, 97.0% with the active-member flag, and a mean of 1.54 products. France is the recorded source category for 50.3% of the group, Spain for 25.5%, and Germany for 24.2%. Median estimated salary is 100,240 and median balance is 95,984 in an unspecified source currency. These numbers describe the static extract only: they are not income, wealth, contribution, CLV, or evidence of a product need.

**What the simulated six months show.** Linh opens the app 67.17 days in six months, or about 11.2 days in a typical simulated month. Her group records 4.83 simulated inflow months and 41.71 transactions in six months. The main synthetic app activity is `transfers_and_bill_pay` (26,472 customer-months); `balance_and_card_controls` appears in another 4,308 customer-months. Failed pay-

ments average 0.25 per customer over six months and unresolved-case share is 7.28%. This is a pattern of frequent digital self-service, not proof of satisfaction or loyalty.

**Why it matters.** The static historical label rate is lower than the other segment at 14.21% (751/5,285), but the group is large. A poorly designed transfer, bill-payment, or card-control flow could inconvenience many customers before it emerges as a complaint. The operational hypothesis is therefore to protect routine completion rather than to send a blanket incentive.

Table 6: Linh persona: evidence, hypothesis, and control [SYNTHETIC_BEHAVIOR + OBSERVED_STATIC_DATA + ASSUMPTION_SCENARIO].

| Dimension | Operational interpretation |
|---|---|
| Strength | An established simulated self-service routine makes contextual help feasible. |
| Weakness | Friction in a frequently used task can damage the experience at scale. |
| Opportunity | Test a task-specific shortcut, clear processing status, and one-step escalation to Care. |
| Threat | Irrelevant sales outreach can create contact fatigue or distract from a system fault. |
| Need to test | Does reducing routine-task friction retain more customers than a transparent, opt-in explanation of existing fees and benefits? |
| Treatment comparison | Digital Concierge versus Value Review; neither arm may use an automatic incentive or default cross-sell. |
| Primary outcome | Incremental 90-day retention; leading indicators are task completion, failed-task-to-support rate, repeat support contact, opt-out, and complaint. |
| Falsification | No positive retention effect with a lower 95% CI above zero, or guardrails worsen versus the comparator. |

### 4.3.2 Persona B — Minh: thin digital relationship, recovery first

**Who Minh represents.** Minh represents 4,715 customers (47.15% of the portfolio). The static profile is not materially defined by wealth: mean age is 38.2, mean credit score 647.5, 46.6% are recorded female, mean product count is 1.52, median estimated salary is 100,138, and median balance is 98,438 in the same unspecified source currency. Only 0.5% carry the active-member flag. Recorded country categories are 50.0% France, 26.1% Germany, and 23.9% Spain. The contrast with Linh shows why balance or estimated salary must not stand in for relationship health.

**What the simulated six months show.** Minh opens the app 24.42 days in six months, or about 4.1 days in a typical simulated month; simulated inflow appears in 1.35 months and transaction count is 26.35. The group primarily reaches `account_status_check` (28,134 customer-months) and rarely reaches transfer or bill-payment tasks. Failed payments average 0.64 in six months and unresolved-case share is 20.91%. The historical static label rate is 27.27% (1,286/4,715). These measures identify a thin simulated relationship and a priority for investigation; they do not diagnose motive, financial vulnerability, or a verified reason for exit.

**Why it matters.** The correct first question is not how to sell more. It is whether the customer has a service problem that deserves resolution, or whether a simple, consented re-entry journey can restore a useful routine. An unresolved case overrides marketing eligibility. The group is never described as

a bad customer; it is a group for which the bank has not demonstrated a stable digital routine.

Table 7: Minh persona: evidence, hypothesis, and control [SYNTHETIC_BEHAVIOR + OBSERVED_STATIC_DATA + ASSUMPTION_SCENARIO].

| Dimension | Operational interpretation |
|---|---|
| Strength | The relationship still exists; an appropriate service or reactivation intervention can be tested. |
| Weakness | The simulated routine is sparse and service-friction indicators are higher. |
| Opportunity | Compare relationship repair against a narrowly scoped return-to-use journey. |
| Threat | A promotion sent before a case is resolved can increase distrust, complaints, and opt-outs. |
| Need to test | Is proactive service recovery more effective than a 14-day guided return to one useful action? |
| Treatment comparison | Service Recovery versus Reactivation Coach, with no cash incentive in the first pilot. |
| Primary outcome | Incremental 90-day active-month rate and retention; leading indicators are SLA closure, re-open, app-active days, failed payment, opt-out, and complaint. |
| Falsification | No positive lower 95% CI for effect, policy breach, or deterioration in complaint, repeat-contact, or failed-payment guardrails. |

## 4.4 Static customer lenses: useful overlays, not extra clusters

The project also retains three static lenses for analysis planning. They are not K-Means segments, treatment rules, or value tiers. Their role is to pre-specify aggregate reporting strata, capacity checks, and fairness review after a valid experiment exists. Table 8 reports the current source-extract cross-tabs. The large differences merit investigation but cannot be read as causal mechanisms or future individual probabilities.

Table 8: Static customer lenses, historical-label cross-tabs [OBSERVED_STATIC_DATA].

| Lens | Group | Customers | Historical label rate | Proper decision use |
|---|---|---|---|---|
| Relationship | Recorded not-active member | 4,849 (48.5%) | 26.9% [25.6, 28.1] | Pre-specify an outcome-analysis stratum; do not auto-contact. |
| Relationship | Recorded active member | 5,151 (51.5%) | 14.3% [13.3, 15.3] | Comparator stratum for pilot reporting. |
| Product depth | 1 product | 5,084 (50.8%) | 27.7% [26.5, 29.0] | Test a product-education hypothesis only through randomization. |
| Product depth | 2 products | 4,590 (45.9%) | 7.6% [6.9, 8.4] | Descriptive reference; not proof of protection from churn. |
| Product depth | 3 products | 266 (2.7%) | 82.7% [77.7, 86.8] | Small group; review service/product context, not an automatic |

## 4.5 Segment cards: decision use, not invented personas

The segment cards in Table 9 convert the evidence into testable service propositions. Names describe the measured pattern rather than assigning motives. Both cards include a suppression rule because a segment alone cannot override customer protection. The example treatments are hypotheses for a future pilot; they are not offers that have been delivered or measured.

Table 9: Segment cards and differentiated decisions [SYNTHETIC_BEHAVIOR + OBSERVED_STATIC_DATA + ASSUMPTION_SCENARIO].

| Field | High digital routine | Lower digital relationship |
|---|---|---|
| Evidence-based name | Customers with an established simulated digital routine | Customers with a thinner simulated digital relationship |
| Scale and historical-label overlay | 5,285 customers; 14.21% static historical label rate | 4,715 customers; 27.27% static historical label rate |
| What distinguishes the group | Frequent app-active days, recurring inflow months, and higher transaction count in the simulation | Fewer app-active days and inflow months; more simulated failed payments and unresolved-case share |
| Operational hypothesis to test | Routine tasks should be completed quickly, with contextual help when friction appears | Service recovery or a simple return-to-use journey may be more appropriate than promotion |
| Proposed comparison | Digital Concierge versus opt-in Value Review | Service Recovery versus a 14-day Reactivation Coach |
| Primary future outcome | 90-day retention, with task completion and repeat support contact as leading indicators | 90-day active-month rate, then retention, with case closure and failed payments as leading indicators |
| Suppression and guardrails | No outreach without consent and within contact cap; pause for complaint/opt-out breach | An unresolved service case routes to care, never marketing; consent breach is an immediate stop |
| What would falsify the card | Stable feature separation disappears with revised generator/real events, or the treatment fails to improve outcome without harms | Same; additionally, service recovery fails to improve closure/re-open or active-month outcome versus control |

## 4.6 From segment to contact queue: decision cube and pilot feasibility

The decision cube in Figure **??** combines behavioural segment with the recorded active-member lens, a simulated service-routing rule, and a simulated inbound-cash-flow scenario. It is deliberately not called a value tier. The value-scenario field is formed from tertiles of simulated inbound flow and supports capacity rehearsal only. It must not determine a price, incentive, or contact priority. The label *eligible for experiment* in this artifact means no unresolved simulated case; actual consent, contact frequency, suitability, and CRM capacity are not available and must be checked before any real eligibility list is created.